LOT PAGER ��� 04-ELECTRONICS-PCBWAY

LOT® PAGER — Electronics & PCBWay

1. Block diagram

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[Qi coil]→[BQ51013B]→5V→[BQ25185]→[Li-Po]→[MAX17048]
                                  └→[TPS62840]→3V3→ ESP32-S3-MINI-1U
ESP32-S3 — SPI — Display (ST7789)
         — DVP — Camera (OV2640/5640) + SCCB
         — I²C — BME688, BMI270, VEML7700, MAX17048
         — GPIO — Button, LED, buzzer/LRA
         — u.FL — external 2.4 GHz antenna on non-metal window
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2. PCB spec (PCBWay quote sheet)

Item	Value
Outline	38 × 38 mm with corner R3 (1 mm clearance to steel)
Layers	4 (SIG/GND/PWR/SIG)
Thickness	**0.8 mm** (not standard 1.6 — z-height)
Min trace/space	0.1 / 0.1 mm (4/4 mil)
Finish	ENIG
Via	0.2/0.4 mm, filled/capped under BGA/WCSP
Copper	1 oz outer, 0.5 oz inner
Impedance	50 Ω on antenna and camera lines, controlled
Colour	Black/matte
Assembly	Both-side SMT, turnkey (PCBWay sources parts) or consigned
Stencil / test	Yes; flying probe
Quantities	Proto 5 → EVT 10 → DVT 25 → run 100 (+10 %)

Feasibility: a flex-rigid is better for v1 (display + coil + camera on flex fold). PCBWay quotes flex and rigid-flex; ask for stack-up review for 0.8 mm 4-layer.

3. PCBWay workflow

1. Design in KiCad 8 (open format, produces Gerbers + BOM + CPL).
2. Upload Gerbers at https://www.pcbway.com/orderonline.aspx for fab quote.

3. Use PCBWay Assembly for BOM sourcing (https://www.pcbway.com/pcb-assembly.html); expect 2–3 line items to be “unavailable” → consign those (BQ51013B, BME688 shortages are common).
4. Add the CNC stainless job at https://www.pcbway.com/cnc-machining-quote.html (STEP + drawing + polish spec).
5. Request DFM report before paying.

4. Design rules / checklist

- Keep camera DVP traces < 50 mm, matched ±5 mm.
- Decouple ESP32-S3 per datasheet (3V3 ≥ 10 µF + 100 nF on each supply pin); 500 mA peak on TX.
- Qi coil keep-out: no copper on layers under the coil; ferrite shield between coil and battery.
- Battery: protection IC, thermistor on pack, never charge < 0 °C.
- ESD (TPD4E05U06) on button and any exposed pad.
- Test points for UART0, SWD-equivalent (USB-JTAG on S3), 3V3, 5V, BAT, I²C.
- Programming: pogo-pin pads (4-pin: 3V3/GND/D+/D−) on PCB edge reachable only with shell open; no USB port on finished unit.